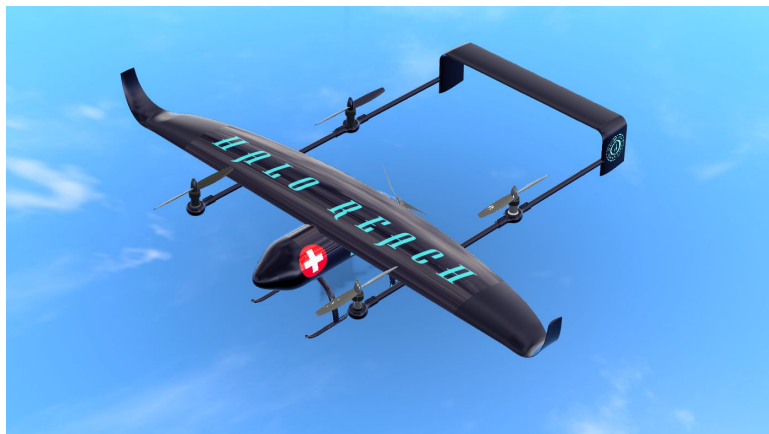


IMPERIAL



HALO REACH: VTOL Forward and Backward Transition Controller Development

UAS for Disaster Relief - AE3 2024



Student: Shun Tang

CID: 02023280

Supervisor: Dr Sophie Armanini

Department: Department of Aeronautics

Course: MEng Aeronautical Engineering

Module: AERO60004 - Group Design Project

Date: 17/06/2024

Academic Year: 2023-2024

Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ
U.K.



Table 1. People involved in HALO REACH - GDP (**student in bold**)

Team code	Team	Names	CID
FATT01	Team Leads	Moreno, Patricia	01857449
		Rashid, Hasan H	02020313
FATT02	Aerodynamics	Alam, Shahriar	02037395
		Dirirsa, Naboth	02026211
		Lawlor, Max F	02026199
		Singh, Parasdeep	02099678
		Homa, Matthew M	02038403
FATT03	Structures	Leng, Jiaxun (Aaron)	02015060
		Wang, Zhiyan	01911209
		Liu, Stephen	02015258
		Chauhan, Shivani	02096006
FATT04	Systems and propulsion	Lin, Yongkai	02029838
		Chursuk, Tharm	02048996
		Wankhade, Divij P	02027906
		Aggarwal, Ishaan K	02030969
		Halim, George	02059571
FATT05	Flight dynamics	Baikady, Thilak S	02016794
		Scott-Brown, Ruairidh	01734080
		Allen, Harry	02028116
		Tang, Shun S	02023280
		Lee, Jeongheon (Edward)	01754474
FATT06	Add-ons	Rosenthal, Adam T	02027024
		Ramteke, Shreya	02079050
		Nakhla, Andrew	02044646

Abstract

In this report, the design of a transition controller for a lift+cruise VTOL is introduced to bridge the takeoff/landing phase and the cruise phase. A longitudinal state space model combining the dynamics of the fixed wing and rotor is developed. Then a transition corridor is designed for forward and backward transition and a set of equilibrium points is hence defined along the corridor. Jacobian Matrices are obtained for the system's A and B matrix, and LQR gain is subsequently calculated respectively. Q and R weight are tuned through a systematic iteration guided by physics intuition and compared with gains from an optimization approach using `fmincon`. Then a code-based simulation is developed for prompt testing and iteration of the controller gains and gain scheduling implementation. Finally, the simulation results demonstrate the tracking performance and transition path of the UAV. Results are then discussed and future improvements are proposed.

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Nomenclature

$\ddot{h}$	Acceleration in y-direction (m/s ²)
m_{total}	Total mass (kg)
N	Normal force (N)
P	Power (W)
x_T	Thrust moment arm (m)
GTOW	Gross Takeoff Weight
HALO	Humanitarian Aid Logistics Operations
LQR	Linear Quadratic Regulator
MI	Merit Index
MPC	Model Predictive Controller
PIDF	Proportional, Integral, Derivative, and Filter
REACH	Rapid Emergency Aid and Communications Hub
RMS	Root Mean Square
UAS	Uncrewed Aerial System
VTOL	Vertical Take-Off and Landing

1 Introduction

VTOL stands for vertical takeoff and landing and followed by extensive evaluation of different VTOL configurations, a lift+cruise VTOL prevails as the most suitable one for the search and rescue mission. Despite some of the disadvantages of the added rotor, such as extra weight, the advantage lies in precisely the combination of flexibility enabled by the ability to VTOL and efficiency enabled by cruising with fixed wing. To bridge the two configurations between the vertical takeoff quadcopter phase and cruise in fixed wing configuration, a control strategy needs to be developed for the transition phase. This report documents the technical details and processes of developing a set of transition controllers for a lift+cruise VTOL Unmanned Aerial Vehicle.

2 Literature Review

As the control of VTOL during transition is still an active research field due to its complex dynamics, numerous literature reviews are conducted to guide the thinking process and compared against each other to inform the optimal approach realistically within the 5-week timeframe. This section reviews the key strategies for transition control.

2.1 Control Strategies for VTOL

- **PID controller:** The use of PID controllers in VTOL UAVs for managing transitions between lift and cruise modes is well-documented. [1] describe a three-loop PID control system to manage transitions in a fixed-wing VTOL UAV, emphasizing the need for stability during the transition phases. Similarly, [2] discusses the decoupled PID control system for attitude and velocity control in VTOL UAVs, highlighting its effectiveness in maintaining stability during different flight phases including hover, transition, and cruise. Moreover, [3] integrate PID control with gain scheduling to handle the dynamic changes during transitions, thereby improving the reliability of the system.
- **MPC controller:** Model Predictive Control (MPC) has emerged as a robust method for managing the complex dynamics of VTOL UAVs during transitions. [4] developed an MPC for a Tail-Sitter VTOL UAV, optimizing the control strategy for hover and transition phases. [5] propose a unified MPC strategy for seamless mode transitioning in tilt-rotor VTOL UAVs, demonstrating its efficacy in handling the varying aerodynamic and thrust characteristics during flight transitions. [6] further explores MPC-based longitudinal control designs specifically for transition flights, highlighting improvements in control precision and stability.
- **LQR controller:** Linear Quadratic Regulator controllers are widely used for their optimal control properties in VTOL UAVs. [7] applied LQR control to ensure smooth transitions in VTOL aircraft, focusing on robustness and uniformity across different flight regimes. [8] designed a discrete-time LQR controller to manage the transition from hover to cruise, addressing the aerodynamic complexities involved. [9] implemented a gain-scheduled LQR controller for transitional flight control, ensuring stability and performance across varying flight conditions.

Overall, LQR controller prevails to be the most suitable strategy due to its optimal control approach and relatively low computational complexity for real-time application as compared to MPC. It has the advantage over PID of handling MIMO system, for which PID requires manual tuning/pole placement of multiple control loops. Although MPC offers superior performance in handling non-linear dynamics, it increased computational complexity and at the same time, it is evaluated to be overly ambitious to develop an MPC transition controller within the 5 weeks timeframe.

2.2 Transition Corridor and Differential Flatness Approach

The concept of a transition corridor is crucial in ensuring safe and efficient transitions for VTOL UAVs. [10] explored differential flatness-based control methods for VTOL UAVs, emphasizing the

importance of defining trim conditions and control inputs for stable transitions. [11] used differential flatness properties to generate aerobatic trajectories for VTOL aircraft, ensuring precise control during aggressive maneuvers. [12] applied differential flatness to periodic control of UAVs, demonstrating its effectiveness in managing transitions with high precision.

2.3 Gain Scheduling for VTOL UAVs

Gain scheduling is another technique in handling the non-linear dynamics of VTOL UAVs during transitions. [13] developed a gain scheduling PID controller for a quad-rotor UAV, enhancing the system's adaptability to changing flight conditions. [14] combined dynamic inversion with gain scheduling for autonomous aerial vehicles, highlighting improved control accuracy across multiple flight stages. [15] introduced a fuzzy gain-scheduling mechanism to adjust PID controllers for UAV position and altitude stabilization, demonstrating significant improvements in control performance.

2.4 Summary

LQR is chosen as the control strategy and since it is designed to handle linear systems, multiple linearisation points need to be identified for gain scheduling. LQR also requires an accurate dynamics model, therefore a full state space system of the VTOL needs to be derived. Inspired by the differential flatness approach, a transition corridor will be manually designed to reduce energy consumption and subsequently, it will define the equilibrium points to linearize the system.

3 State Space Model Derivation

From this section onward, one assumption is made throughout: longitudinal dynamics are decoupled from lateral dynamics. The assumption greatly simplified the complexity of the state space equations and the assumption can be made as an initial attempt because the transition phase is largely concerned with longitudinal velocity and pitch. It is a valid assumption until lateral gust disturbance is considered.

The motivation for a nonlinear longitudinal state space model is: firstly, to capture the complex dynamics to enable the design of an effective model-based controller; secondly, to be used for a nonlinear simulation model. A free-body diagram is drawn as shown in Figure 3 to illustrate the force and moment balance on the UAV. Note that the body frame is deployed as the frame of reference with positive x pointing leftwards, positive z pointing downwards, and positive moment clockwise.

The states of the system are given as $\mathbf{x}$ and the inputs $\mathbf{u}$ are given as:

$$\mathbf{x} = [u, w, q, \theta]^T \quad \mathbf{u} = [\delta_e, T_p, T_{MR,f}, T_{MR,b}]^T$$

where u and w are horizontal and vertical velocity respectively and q and θ are pitch rate and pitch angle (with respect to the vertical axis on earth frame). Thus the state space equation is derived as:

$$\begin{aligned} \dot{u} &= \frac{X + T - mg \sin \theta}{m} & \dot{q} &= \frac{\tau_y + M}{I_{yy}} \\ \dot{w} &= \frac{Z - T_{MR} + mg \cos \theta}{m} & \dot{\theta} &= q \end{aligned}$$

where the total moments and the total propeller thrust are given as follows:

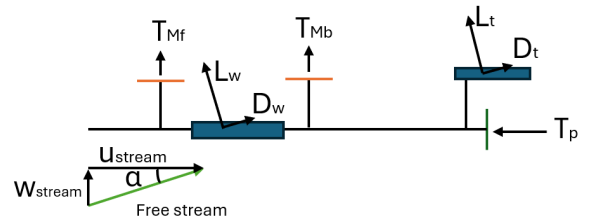


Figure 3. Free body diagram for the UAV

$$\tau_y = \sum_{i=f}^b r_{i,x} T_{MR,i} \quad T_{MR} = \sum_{i=f}^b T_{MR,i}$$

with X , Z , M referring to aerodynamics drag, lift, and moment respectively while T , T_{MR} , τ_y refer to propulsive thrust, total lifting thrust, and overall moment induced by the rotors.

4 Aerodynamics Model Used in Equation of Motion (EoM)

The aerodynamic model is subsequently derived for a large range of angle of attack. It is necessary because in the transition phase, due to a low forward velocity, any disturbances or comparable vertical velocity can subject the UAV to a large angle of attack or negative angle of attack. The EoM and aerodynamic model were derived prior to any CFD results which can be found in [16] therefore flat plate post-stall dynamics was deployed incorporating the effects of wing stall and nonlinearity and drag is subsequently derived using induced drag and parasitic drag equation.

$$\begin{aligned} X &= \bar{q}S(-c_D \cos \alpha + c_L \sin \alpha) \\ Z &= \bar{q}S(-c_D \sin \alpha - c_L \cos \alpha) \end{aligned}$$

The dynamic pressure, aerodynamic angles, and airspeed are defined as:

$$V = \sqrt{u^2 + w^2} \quad \bar{q} = \frac{1}{2}\rho V^2 \quad \alpha = \tan^{-1}\left(\frac{w}{u}\right)$$

4.1 Lift Coefficient

The lift force F_{lift} can be expressed as:

$$F_{\text{lift}} = \bar{q}S \left[C_L(\alpha) + C_{Lq} \frac{c}{2V} q + C_{L\delta_e} \delta_e \right]$$

where S is the wing area.

To account for wing stall, the lift coefficient C_L is modeled as a nonlinear function of α . For angles of attack beyond the onset of stall, the wing lift can be approximated with a flat plate [17]. The lift coefficient can be approximated by blending a linear function of α with the lift coefficient of a flat plate[17]:

$$\begin{aligned} C_L(\alpha) &= (1 - \sigma(\alpha))(C_{L0} + C_{L\alpha}\alpha) + \sigma(\alpha) \\ &\quad [2 \cdot \text{sign}(\alpha) \sin^2(\alpha) \cos(\alpha)] \end{aligned}$$

where:

- $\sigma(\alpha)$ is a blending function defined as:

$$\sigma(\alpha) = \frac{1 + e^{-M(\alpha-\alpha_0)} + e^{M(\alpha+\alpha_0)}}{(1 + e^{-M(\alpha-\alpha_0)})(1 + e^{M(\alpha+\alpha_0)})}$$

- M and α_0 are positive constants and after some iterations, 18 and 0.243 rad are used respectively to achieve a close fit to the actual stall [16].

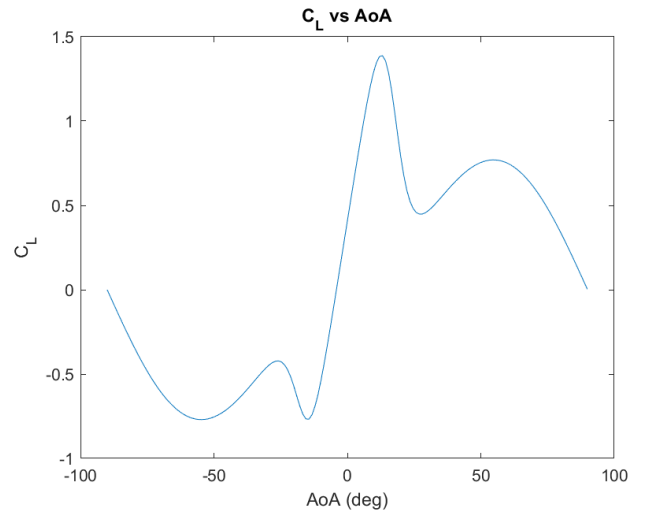


Figure 4. Graph of C_L vs AoA

- C_{L0} and $C_{L\alpha}$ are 0.41 and 5.725 respectively [16];

With the final input from the aerodynamics team [16], the lift coefficient vs AoA graph is plotted as shown in Figure 4. The effect for q is assumed to be negligible for the wing since the wing is very close to the center of gravity as compared to the tail therefore the effect of q is only captured in tail dynamics. The overall lift coefficient is shown below:

$$C_L(\alpha, q, \delta_e) = (1 - \sigma(\alpha))(C_{L0} + C_{L\alpha}\alpha) + \sigma(\alpha) [2 \cdot \text{sign}(\alpha) \sin^2(\alpha) \cos(\alpha)] + \left(\tan^{-1} \left(\frac{q \cdot l_h + w}{u} \right) \right) C_{L\alpha h} + C_{L\delta_e} \delta_e;$$

4.2 Drag Coefficient

The drag force F_{drag} can be expressed as:

$$F_{\text{drag}} = \bar{q}S [C_D(\alpha) + C_{D\delta_e}\delta_e]$$

The drag coefficient C_D is also a nonlinear function of the angle of attack and has two main components: induced drag and parasitic drag. The parasitic drag is roughly constant, denoted by C_{Dp} . The induced drag is proportional to the square of the lift force[17]. This approach not only approximates the drag in the linear region well but also extends the validity to a negative angle of attack as shown in Figure 5. Combining these, the drag coefficient is given by:

$$C_D(\alpha) = C_{D0} + \frac{(C_{L0} + C_{L\alpha}\alpha)^2}{\pi e AR}$$

where:

- e is the Oswald efficiency factor where 0.8 is used [16]
- C_{D0} is the parasitic drag coefficient where 0.0820 is used [16].

4.3 Moment Coefficient

Moment coefficients require CFD or wind tunnel measurement [17] so an empirical value from VTOL UAV with the same configuration [18] was adopted and applied. The expression can be found below:

$$C_M = C_{M0} + C_{M\alpha}\alpha + C_{M\hat{q}}\hat{q} + C_{M\delta_e}\delta_e$$

where

$$C_{M0} = 0.0950, \quad C_{M\alpha} = -1.495$$

$$C_{M\hat{q}} = -13.140, \quad C_{M\delta_e} = -1.675$$

5 Define Transition Corridor and Equilibrium Points

In order to identify the equilibrium points, a fliaible transition corridor needs to be defined. The transition corridor for both the forward and backward transition of the VTOL UAV is defined in two phases. The goal is to design a state evolution and determine a series of equilibrium points throughout the transition, and lastly linearize the system around these points to find the LQR gains. There are a few assumptions made in this section:

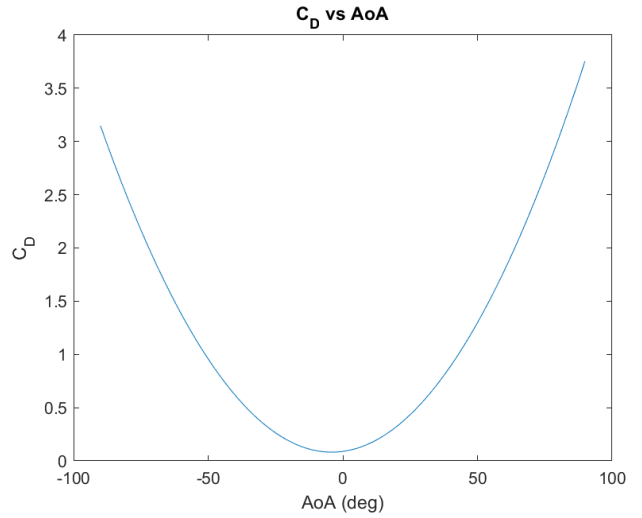


Figure 5. Graph of C_D vs AoA

- The pitch rate is assumed to be zero for equilibrium condition so that $\dot{\theta}$ remains zero by definition. Therefore there are effectively three equations in the state space equations but four unknown control inputs, which results in one redundancy.
- Following the above assumption, the elevator deflection angle is chosen to be the redundant control input and are therefore kept constant. The assumption made subsequently is that the forward velocity throughout the transition period is small and the elevator deflection is ineffective in terms of achieving moment balance. Hence, the elevator deflection is kept at the value in the trim point at the end of the transition period.
- The last assumption made is that the equilibrium condition is more sensitive to the deviation in pitch angle than the deviation in velocity. This is because velocity only changes aerodynamic force which is insignificant in transition while pitch angle is linked to all forcing components. This assumption is validated in the later stage.

5.1 Definition of the range of transition

In this section, forward transition is defined as the phase starting from hover condition where all the states are zero to multirotors switched off where all the lift is generated by the wing and similarly, backward transition is defined as the phase from end of cruise with zero thrust from multirotor to zero velocity hover.

First of all, the trim condition at the end of the transition needs to be defined. Based on the definition of the end of the transition, the trim condition is assumed to be the state when the aircraft operates at stall angle, θ_{stall} of 13.94 deg [16] with zero pitch rate and multirotor input. This defines three simultaneous equations with three unknowns. The state space equation

$$\dot{x} = f(x, u) = 0$$

is substituted with $u = V_{\text{cruise}} \cos(\theta_{\text{stall}})$, $w = V_{\text{cruise}} \sin(\theta_{\text{stall}})$, $q = 0$, $\theta = 0.243$, $T_{MRf} = T_{MRb} = 0$, to define an equation solving for trim condition immediately after transition as shown in Table 2.

Table 2. Table of trim condition immediately after transition

	u (m/s)	w	q (rad/s)	$\theta(\text{rad})$	$\delta_e(\text{rad})$	T_p (N)	T_{Mf}	T_{Mb}
Trim Condition	19.4	4.80	0	0.243	-0.157	278	0	0

5.2 Forward Transition Corridor

It is crucial to identify a stable transition state evolution such that the transition is flialle. Although differential flatness approaches can enable an exact definition of transition corridor [10] which subsequently allows for trajectory optimization, it is too ambitious to explore in five weeks and decided that the method can be deployed in the next iteration for trajectory optimization. However, the idea of energy minimization is inspired by it and some requirements are subsequently defined:

- The pitch angle should be kept zero during the acceleration phase when multirotor thrust is significant to prevent the adverse horizontal component of the thrust due to pitching up.
- The attitude should be kept relatively constant to prevent sudden drop which is crucial to ensure the integrity of fragile medical items to be delivered to the disaster scene.

Therefore two phases are defined subsequently

1. Duration: 20 seconds. Transition in u from zero to V_{cruise} and keeping θ zero.
2. Duration: 60 seconds. Transition in θ from zero to the stall angle θ_{stall} while maintaining forward velocity in earth frame at V_{cruise} .

The equilibrium transition velocity $u_{\text{transition}}$ is incrementally sampled from a small value to V_{cruise} in steps of $u_{\text{step}} = 1(m/s)$. The corresponding transition angles $\theta_{\text{transition}}$ are incremented in steps of $\theta_{\text{step}} = 0.5(deg)$. A total of 46 equilibrium points are obtained. For each transition point, the MATLAB code solves for the equilibrium values of the control inputs δ_t , δ_{MRf} , and δ_{MRb} . The evolution of the states and control inputs are shown in Figure 8 and 9:

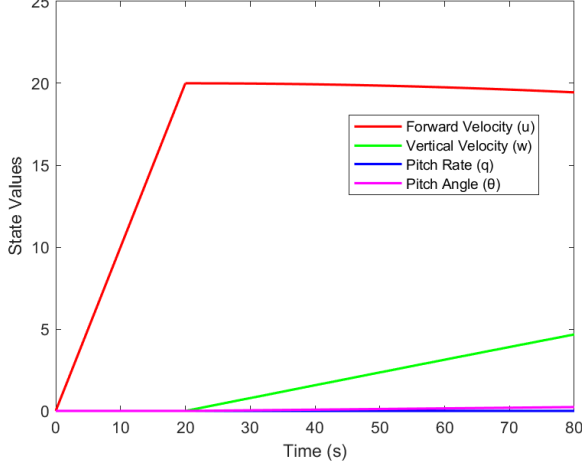


Figure 6. Graph of the Equilibrium Points in the Transition Corridor

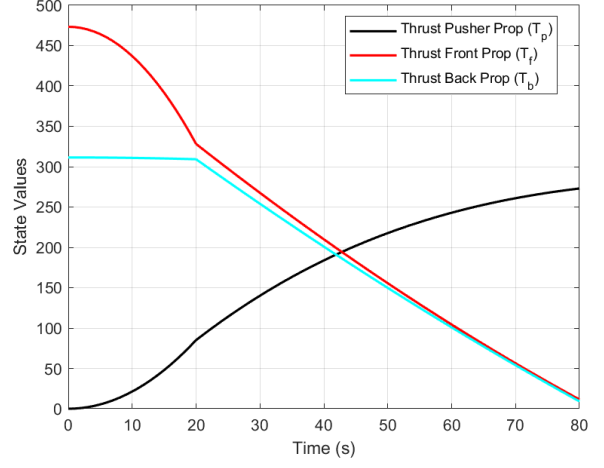


Figure 7. Graph of the Control Input in the Transition Corridor

The equilibrium points are then used to compute the system matrices **A** and **B** by evaluating the Jacobians at each point. LQR gains **K** are subsequently calculated for each equilibrium point to stabilize the system during the transition.

5.3 Backward Transition Corridor

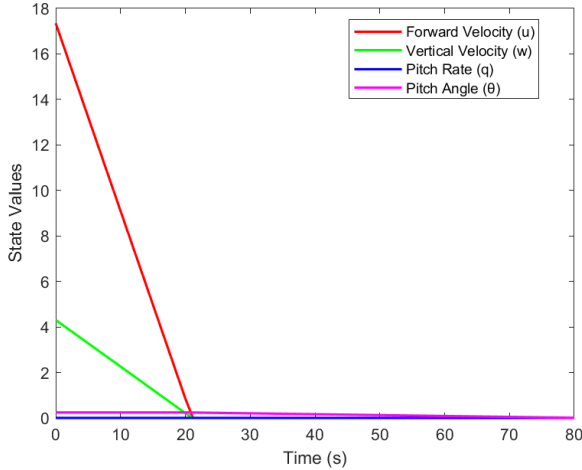


Figure 8. Graph of the Equilibrium Points in the Backward Transition Corridor

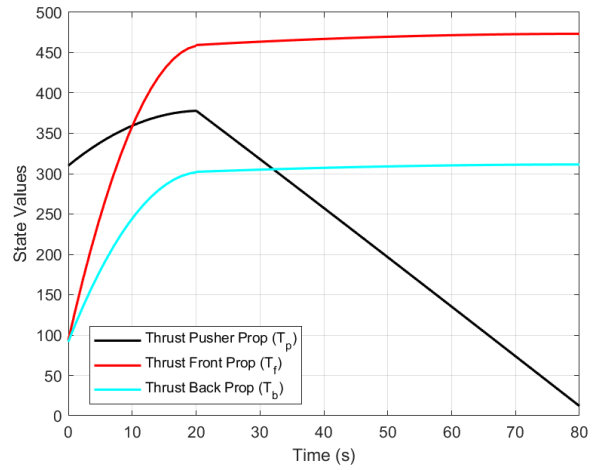


Figure 9. Graph of the Control Input in the Backward Transition Corridor

Similarly, the backward transition is derived however under a different set of considerations:

- It is impractical to slow down the UAV to zero velocity by drag alone because it would take 1000 seconds estimated by [19]. Therefore the backward transition controller needs to maintain the UAV at θ_{stall} to slow down the velocity with adverse horizontal components from the thrust vector.
- The altitude should be kept constant as well.

Therefore two phases are defined subsequently

1. Duration: 20 seconds. Transition in u from V_{cruise} to zero and keeping θ at θ_{stall} .
2. Duration: 60 seconds. Transition in θ from stall angle to zero while maintaining zero velocity.

Similarly, the equilibrium points are sampled, and again a total of 46 equilibrium points are obtained and the evolution of the states and control inputs are shown in Figure 8 and 9.

6 Controller Development

6.1 Simulation Setup

A code-based simulation is set up for quick testing, validate the control strategies and evaluate the performance of the UAV during the transition phase. A for-loop at 0.001s timestep is deployed. The MATLAB code iterates through the simulation time steps, tracking a full state trajectory which is the same as the transition corridor, applying the gain scheduled LQR control inputs (which will be introduced later), and updating the state of the UAV. The results of the simulation, including the state trajectories and control inputs, are recorded and analyzed to evaluate the performance and stability of the control system.

The results of the simulation, including the state trajectories and control inputs, are recorded and analyzed to evaluate the performance and stability of the control system.

6.2 LQR Tuning

A brief explanation of the LQR controller and the role and significance of Q and R matrices can be found in Appendix A. There are a total of 4 sets of Q and R values chosen and each sets of Q and R are tuned for different phases in the forward and backward transition because each phase corresponds to a different physical scenario and would have different tradeoffs among the states and control inputs.

6.2.1 Systematic Tuning Strategy for Q and R Matrix

Although the tuning of Q and R is an iterative process, physical understanding of the process can inform a more systematic tuning approach as the dynamics of some states are inherently coupled:

- The pitch rate and pitch angle are coupled because in order to keep pitch angle constant, a pitch rate will result from corrective action, therefore pitch rate should be less 'stiff' or penalised less in order to avoid excessive oscillation which could lead to instability in actual practice.
- The horizontal velocity u and vertical velocity w are also coupled as an increase/decrease in horizontal velocity will also increase/decrease instantaneous lift which will result in vertical acceleration. In transition, the altitude needs to be kept constant therefore w should be 'stiffer' than u to avoid oscillation and maintain stability throughout transition flight.
- The front propeller and back propeller should be given the same R weights because they are actively adjusting the thrust to keep force and moment balance.
- The R weight for the elevator deflection angle has to be penalised heavily because it is ineffective in generating pitching moments and is therefore a waste of energy. So it should be altered minimally.

With the above physics intuition in mind, an iterative process is applied as following:

1. Adjust Q :

- Increase the diagonal elements of Q corresponding to u and θ that need tighter regulation.
- Decrease the diagonal elements of Q for w and q that can tolerate larger deviations.

- Recompute the LQR gain matrix K and simulate the response.
- Repeat this step until the state deviations are within acceptable limits.

2. Adjust R :

- Increase the diagonal elements of R to reduce control effort and achieve smoother control actions.
- Decrease the diagonal elements of R for the more aggressive control actions i.e. the propellers to meet performance specifications.
- Recompute the LQR gain matrix K and simulate the response.
- Repeat this step until the control effort is balanced with the desired performance.

3. Iterate and Converge:

- Iterate between adjusting Q and R until the desired balance between state regulation and control effort is achieved.
- Validate the final tuning by performing extensive simulations.

The final set of Q and R matrices are presented in Table 3.

Table 3. Table of Q and R weights for different phases in transition

Q/R Weight	u	w	q	θ	δ_e	T_p	T_{MRf}	T_{MRb}
Phase 1 forward	1	5	5	10	3	0.08	0.09	0.09
Phase 2 forward	2	4	1	8	3	0.15	0.09	0.09
Phase 1 backward	2	4	5	10	3	10	0.09	0.09
Phase 2 backward	1	1	0.8	5	3	0.15	0.09	0.09

6.2.2 Q and R Matrix Optimisation

To reach an optimal performance in terms of minimal settling time and overshoot, an optimization technique is experimented with to define the optimal Q and R matrix. A Matlab script is written that examines the step response of the feedback system at one equilibrium point in phase 1 of forward flight. The closed-loop feedback system

$$\dot{x} = (A - BK)x + Bu \quad y = Cx$$

where A and B matrix is the Jacobian matrix of the state space system while C is an identity matrix outputting the current state for feedback and K is the gain matrix based on a selection of Q and R that needs to be optimised. The tracking performance of u is set to be optimised. Hence, the cost function is defined as the sum of the settling time and overshoot for the step response of u to the four inputs. Subsequently, an `fmincon` function for global search is deployed to find the optimal Q and R matrices. Generalizing the optimal Q and R weights for all equilibrium points in phase 1, the tracking performance was compared between the iteratively tuned gains and `fmincon` gains as shown in Figure 10. Counterintuitively, although optimal gains track the trajectory more closely in general, it induces some oscillations around

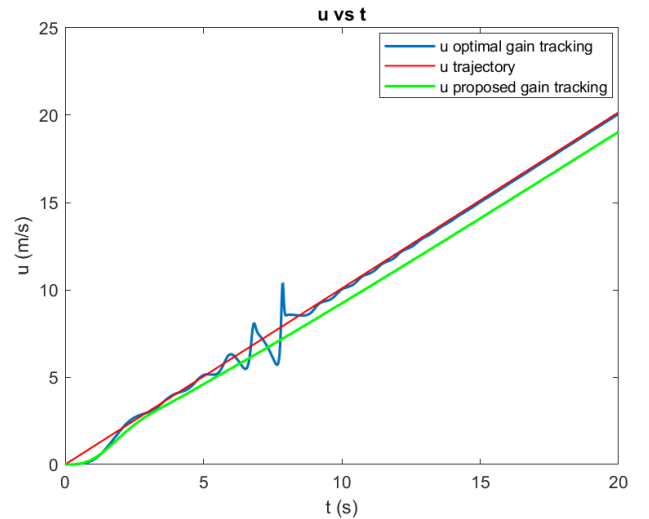


Figure 10. Performance comparison between the proposed gain and optimal gain based on `fmincon`

certain equilibrium points which could destabilize the system in practice. It can be mitigated if such optimization can be performed for all equilibrium points to get a unique set of Q R matrices, which is worth doing when the final controller with a more sophisticated state space model is developed in the later design stage. This experimentation informs a better approach to be taken in the future. This in turn motivates the investigation into the minimum number of equilibrium points for which the dynamic of the system in between points can still be approximated by a linear interpolation. For phase 1 of the transition where only the velocity is changing, the tracking performance of using only 3 equilibrium points is still similar as compared to the 18-point proposed solution as shown in Figure 11. However, the tracking performance of pitch angle, although stable, becomes compromised with fewer equilibrium points as seen in Figure 12.

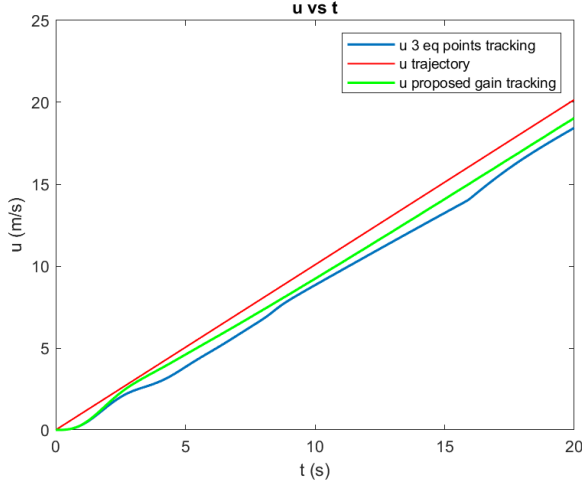


Figure 11. u tracking Performance of 3 equilibrium points controller benchmark against proposed solution

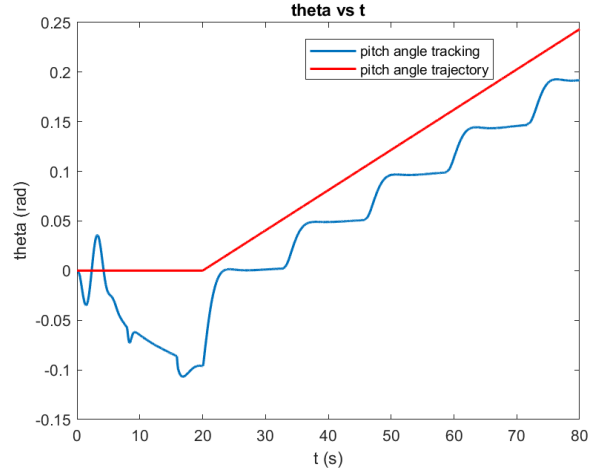


Figure 12. θ tracking Performance of 5 equilibrium points controller benchmark against proposed solution

The velocity tracking is less affected by the number of equilibrium points likely because the dynamic change of equilibrium points along phase 1 of the transition is more linear likely because the lift increased due to the wing and tail are proportional, keeping the moment balanced thus the dynamics are less coupled; while the change in pitch angle induces a highly nonlinear change of the coupled dynamics which results in a staggered tracking response in Figure 12. Therefore, in the next iteration, phase one equilibrium points can be reduced while phase two needs to be handled more carefully with more iterative testing to achieve convergence.

6.3 Gain Scheduling

To ensure stable and efficient control throughout the transition phase, gain scheduling technique is employed to achieve stability by interpolating the gain matrix K in between two equilibrium points to prevent instability due to large deviations from the equilibrium condition. Model-based LQR controller is sensitive to the accuracy of the model, in this context, deviation from the equilibrium points, therefore a large deviation could potentially result in instability and therefore gains scheduling mitigates this effect. The gain scheduling logic adapts the LQR controller gains based on the scheduling parameter of the system by interpolating the gain matrix based on the 'progress' between two equilibrium points. In the acceleration/deceleration phase, u is set to be the scheduling parameter and in pitching up/down phase, θ becomes the scheduling parameter.

7 Simulation Results and Discussion

In this section, trajectory tracking performance is presented.

7.1 Forward Transition

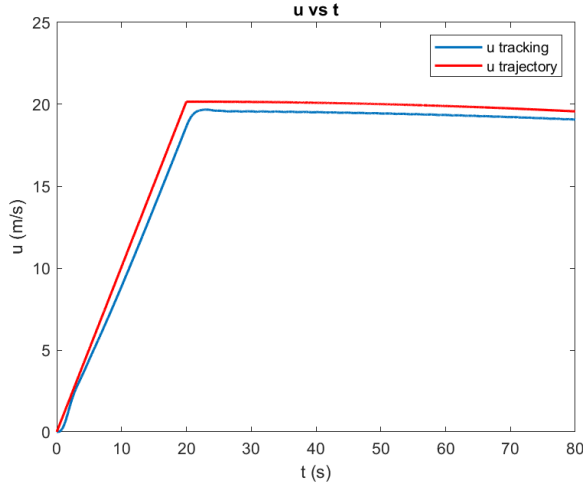


Figure 13. Trajectory Tracking of u

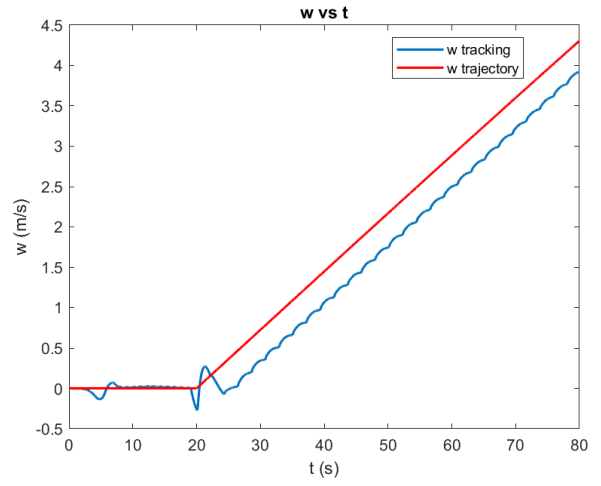


Figure 14. Trajectory Tracking of w

The tracking performance of u and w is shown in Figure 13 and 14. u tracking performance is satisfactory with minimum tracking error of 5%. While phase 1 is tracked accurately, w tracking trajectory in phase 2 deviates constantly from the set trajectory. This is likely because the system has an inherent steady-state error in the tracking of a ramp input and LQR does not have an integral term. The natural solution is to adopt a LQI controller which can eliminate the steady-state error and improve tracking performance.

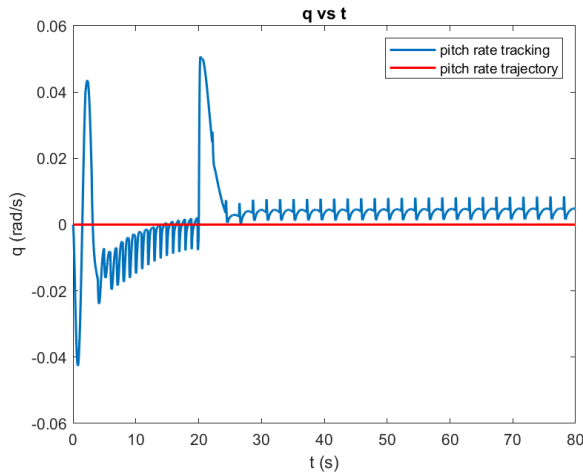


Figure 15. Trajectory Tracking of q

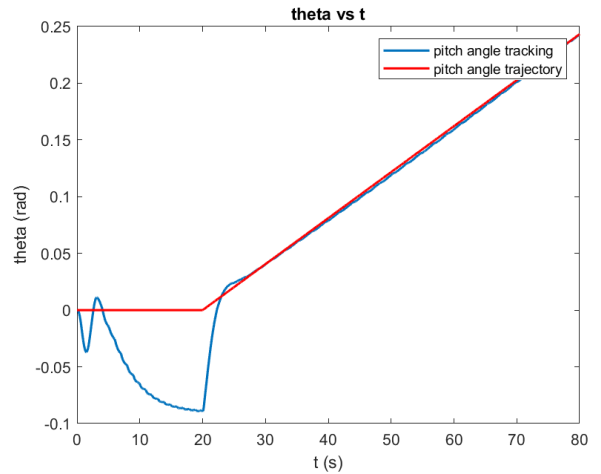


Figure 16. Trajectory Tracking of θ

The tracking performance of q and θ is shown in Figure 15 and 16 and generally the phase 2 tracking performance is excellent but phase 1 exhibits more significant error. The tracking error of θ in phase 1 emerged because of the characteristics of quadcopter dynamics of the UAV as it pitches down to accelerate the UAV to keep track of the u trajectory. As for q , there exists a rapid oscillation in phase 1 and tiny periodic oscillation in phase 2. The larger scale oscillation is to be expected because as the UAV accelerates, more lift is generated from the wing, and the front and back rotors are adjusted rapidly to reduce lift which could result in instantaneous moment and therefore change of pitch rate. As for the phase two periodic oscillation, it likely corresponds to switching equilibrium points and pitching up.

7.2 Backward Transition

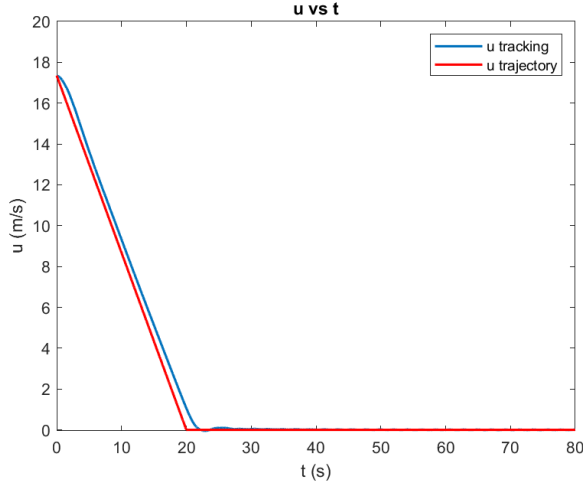


Figure 17. Trajectory Tracking of u

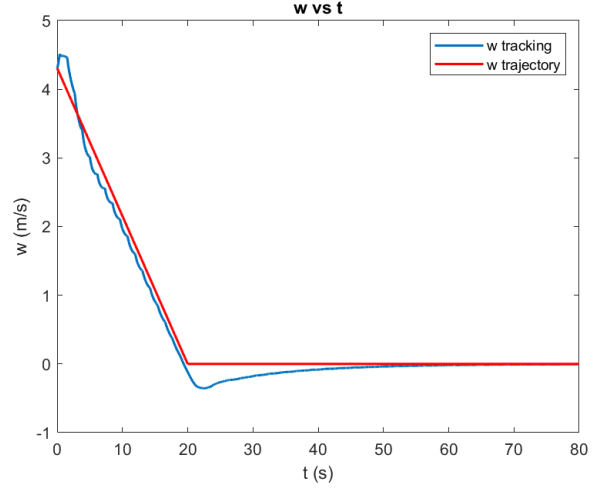


Figure 18. Trajectory Tracking of w

The tracking performance of u and w is shown in Figure 17 and 18 and demonstrates accurate results. The tracking error at the start of each phase is to be expected due to sharp changes in the trajectory.

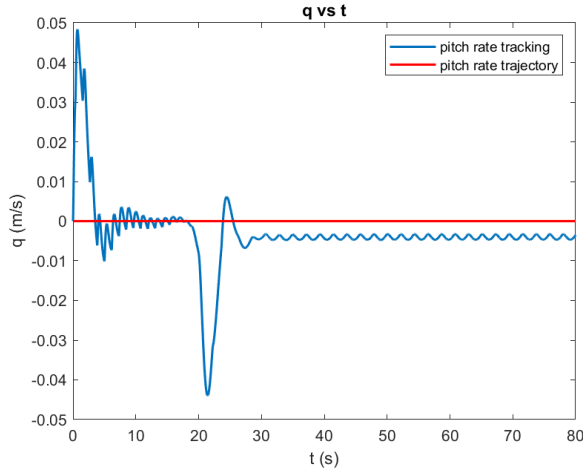


Figure 19. Trajectory Tracking of q

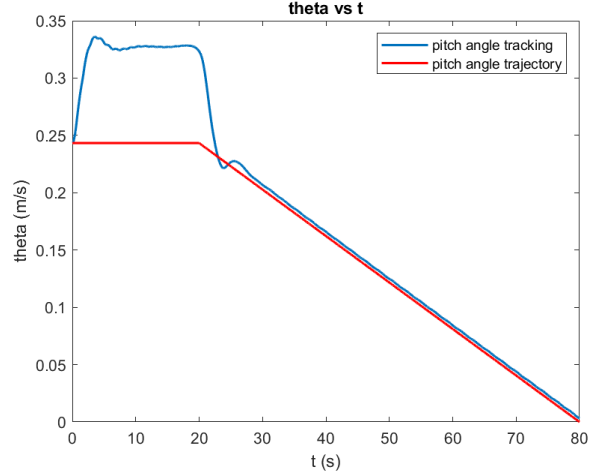


Figure 20. Trajectory Tracking of θ

The tracking performance of q and θ is shown in Figure 19 and 20 and likewise to the forward transition case, error mainly occurs in phase 1. A large constant tracking error of θ in phase 1 emerged again due to the characteristics of quadcopter dynamics of the UAV as it pitches up to match the deceleration required to keep track of the u trajectory. As for q , large oscillations are applied at the start of each phase to initiate the pitch angle change.

7.3 Transition Path

The velocity is converted from the body frame to the earth frame using rotational operators. Then the earth frame velocity is integrated in the 80-second transition period to find the position trajectory and visualise the path. The results are presented below in Figure 21, 22, 23 and 24. As can be observed, the maximum drop in altitude in the forward transition is 16m over the course of 20s and in backward transition, the UAV comes to rest at merely 190m and the altitude increases gradually before entering the hover mode.

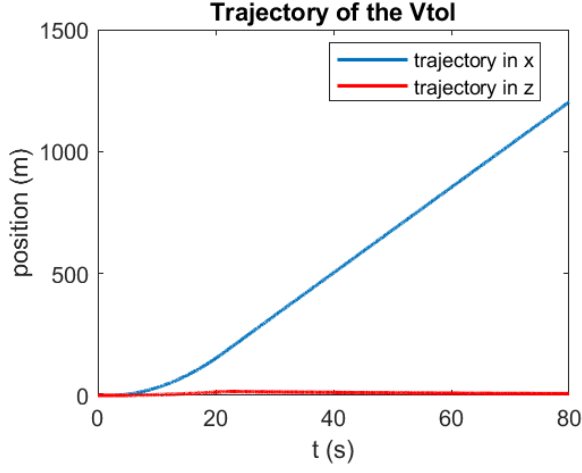


Figure 21. Forward Position Trajectory

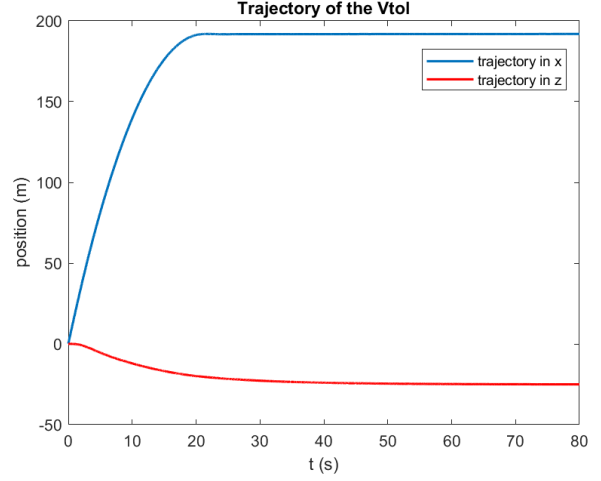


Figure 22. Backward Position Trajectory

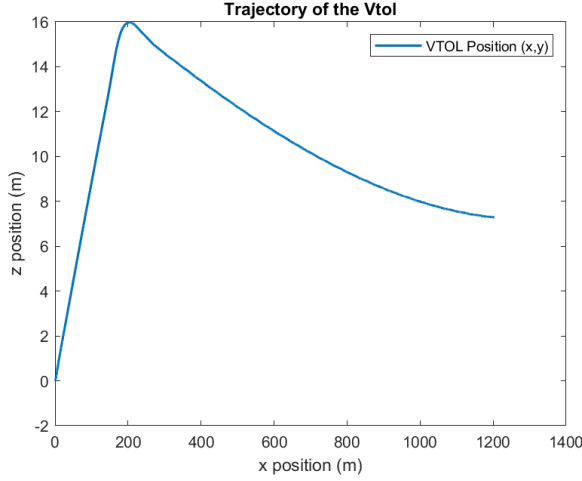


Figure 23. Forward Transition Path

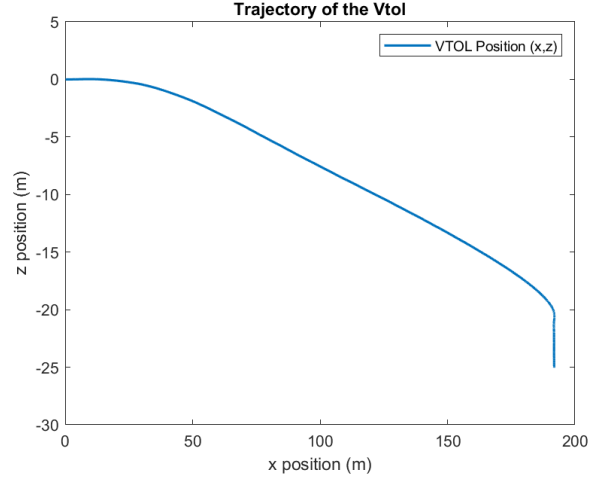


Figure 24. Backward Transition Path

8 Future Work and Improvements

The first iteration of the transition controller has been developed so far and many improvements can be made for future work.

- **Aerodynamics Model:** a more accurate aerodynamic model using curve fitting technique from the result in CFD can be done to replace the current approximate models for C_L , C_D and C_M ; the wake effects of the multirotor on the lift generated by the wing can also be incorporated to get a more accurate aerodynamic model.
- **State Space Model:** a full state space model with both longitudinal and lateral dynamics can be derived so that a full set of gust disturbance analyses can be conducted.
- **Controller:** an optimized trajectory for energy-efficiency can be obtained using a differential flatness approach as a next step to define a more rigorous transition corridor. Then a minimum number of equilibrium points can be investigated and the optimal Q and R matrices optimized for overshoot and setting time can be found using the method as described in Section 6.2.2.
- **Simulation and robustness testing:** after a full state space system is derived, it is worth developing a full simulink-based environment in the time domain with an observer. Disturbances and noise in the sensors can be modeled so that the controllers can be thoroughly tested for robustness and disturbance analysis.

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Appendix

A Significance of Q and R Matrices

LQR is a control strategy used to optimize the performance of a system by minimizing a cost function. The cost function is defined as [3] :

$$J = \int_0^{\infty} (x(t)^T Q x(t) + u(t)^T R u(t)) dt$$

where $x(t)$ is the state vector, $u(t)$ is the control input vector, Q is a symmetric positive semi-definite matrix that penalizes deviations in the state vector, and R is a symmetric positive definite matrix that penalizes the control effort.

The matrices Q and R play crucial roles in the design of the LQR controller:

- Q Matrix: This matrix determines how much penalty is applied to each state changes. Larger values in Q increase the cost of deviating from the desired state, leading to more aggressive control actions to minimize these deviations.
- R Matrix: This matrix determines how much penalty is applied to the control effort. Larger values in R reduce the control input, leading to smoother and more conservative control actions.

Properly tuning Q and R is essential to balance the trade-off between state regulation and control effort, ensuring that the VTOL aircraft transitions smoothly and efficiently between flight modes.